

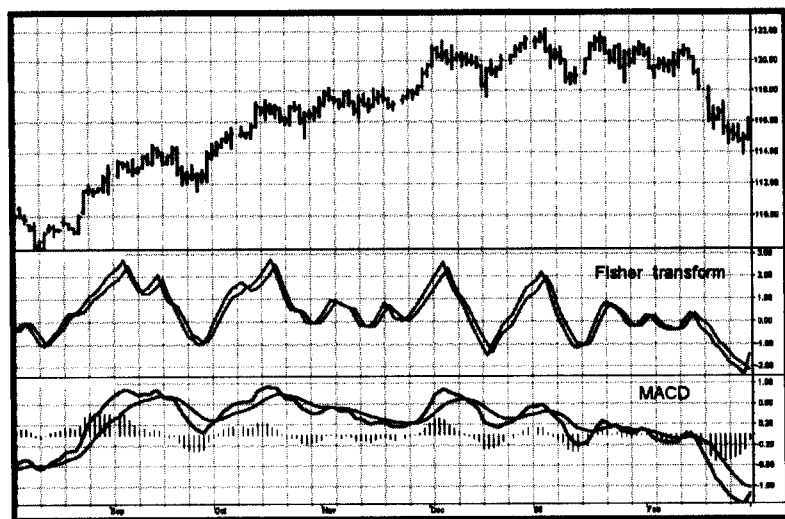


FIGURE 1.9 The Fisher Transform of Normalized Prices Has Very Sharp Turning Points When Compared to Conventional Indicators such as the MACD

KEY POINTS TO REMEMBER

- Prices almost never have a Gaussian, or Normal, probability distribution.
- Statistical measures based on Gaussian probability distributions, such as standard deviations, are in error because the probability distribution assumption underlying the calculation is in error.
- The Fisher transform converts almost any input probability distribution to be nearly a Gaussian probability distribution.
- The Fisher transform, when applied to indicators, provides razor-sharp buy and sell signals.

CHAPTER 2

Trends and Cycles

"That took the wind out of my sails," said Tom disgustedly.

To a trader, Trend Modes and Cycle Modes are synonymous with selection of a trading strategy. In an uptrend the obvious strategy is to buy and hold. Similarly, in a downtrend the strategy is to sell and hold. Conversely, the best strategy in a Cycle Mode is to top-pick and bottom-fish. Traders usually use some variant of moving averages to trade the Trend Mode and some oscillator to trade the Cycle Mode. In either case, the lag induced by the calculations is one of the biggest problems for a trader.

To an analyst, Trend Modes and Cycle Modes are best described by their frequency content. Prices in Trend Modes vary slowly with respect to time. Therefore, Trend Modes disregard high-frequency components and use only the slowly varying low-frequency components. Moving averages are low-pass filters that allow only the low-frequency components to pass to their output, and that is why they are effective for Trend Mode trading. Oscillators are high-pass filters that almost completely disregard the low-frequency components.

I will use these concepts to create a complementary oscillator and moving average. Most important, both the oscillator and the moving average have essentially no lag. The elimination of lag is crucial to the trading indicators and systems developed from them in later chapters. I consider the creation of these zero-lag tools one of the most important developments described in this book. Searching for zero-lag tools has long been the focus of my research, and I have used descriptors such as *Instantaneous Trendline* in previous publications. The techniques I show you in this chapter are entirely new, even if the names are similar.

I will start with the well-known exponential moving average (EMA) to derive an optimum mathematical description of Trend Mode and Cycle Mode components. The equation for an EMA is

$$\text{Output} = \alpha * \text{Input} + (1 - \alpha) * \text{Output}[1] \quad (2.1)$$

Where α is a number less than 1 and greater than 0

In words, this equation means we take a fraction of the current price and add to it the filtered output one bar ago multiplied by the quantity $(1 - \alpha)$. With these coefficients, if the input is unchanging (zero frequency), the output will eventually converge to the input value. That is, this filter has unity gain at zero frequency. We can describe this filter in terms of its transfer response, which is the output divided by its input. By using Z transform notation, we let Z^{-1} denote one bar of lag as a multiplicative operator. Doing this, the transfer response of Equation 2.1 can be solved using algebra as

$$H(z) = \frac{\text{Output}}{\text{Input}} = \frac{\alpha}{1 - (1 - \alpha) * Z^{-1}} \quad (2.2)$$

We can test Equation 2.2 by letting Z^{-1} equal +1 (zero frequency). When we do this, it is easy to see that the numerator is equal to the denominator, and so the gain is unity. The high-frequency attenuation of this filter can be tested at the highest possible frequency, the Nyquist frequency, by letting Z^{-1} equal -1 . Using daily samples, the highest frequency we can analyze is 0.5 cycles per day (a two-bar cycle). This is the Nyquist frequency for daily data. The two-bar cycle attenuation is $[\alpha/(2 - \alpha)]$. The general attenuation response of the EMA as a function of the frequency is shown in Figure 2.1. The period of a cycle component in Figure 2.1 can be calculated as the reciprocal of frequency. For example, a frequency of 0.1 cycles per day corresponds to a 10-bar period for that cycle component.

In principle, all we have to do to create a high-pass filter is subtract the transfer response of the low-pass filter from unity. The logic is that a transfer response of 1 represents all frequencies, and subtracting the low-pass response from it leaves the high-pass response as a residual. However, there is one problem with this approach: The high-frequency attenuation of the low-pass filter of Equation 2.2 is not infinite (i.e., the transfer response is 0) at the Nyquist frequency. A finite high-frequency response in the low-pass filter will lead to a gain error in the transfer response of the high-pass filter. The finite attenuation problem is eliminated by averaging two sequential input samples rather than using only a single input sample. In this case, the transfer response of the averaged-input low-pass filter is

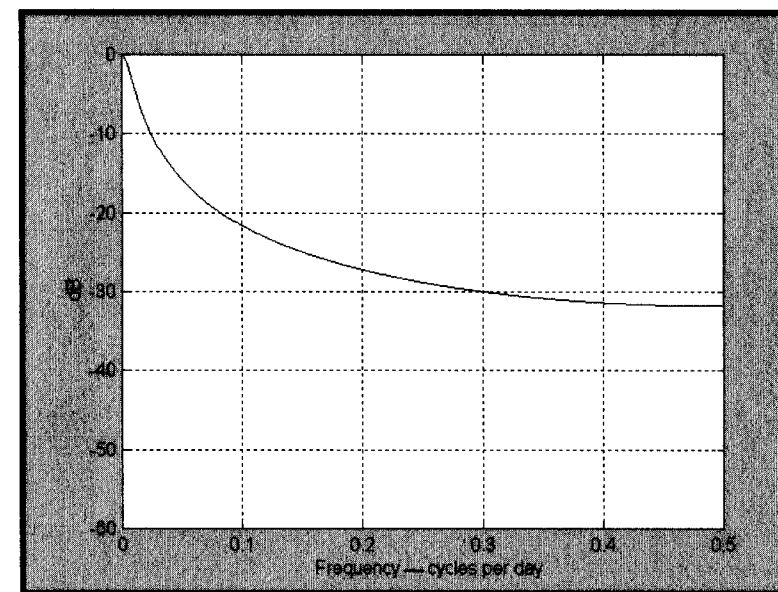


FIGURE 2.1 EMA Frequency Response ($\alpha = 0.05$)

$$H(z) = \frac{\frac{\alpha}{2} * (1 + Z^{-1})}{1 - (1 - \alpha) * Z^{-1}} \quad (2.3)$$

Equation 2.3 guarantees that the transfer response of the low-pass filter will be 0 when $Z^{-1} = -1$. The general frequency response of the averaged-input EMA is shown in Figure 2.2.

The lag of a simple moving average is approximately half the average length. For example, a 21-bar moving average has a lag of 10 bars. The alpha of an equivalent EMA is related to the length of a simple moving average as

$$\alpha = \frac{2}{\text{Length} + 1} \quad (2.4)$$

Using Equation 2.4, an EMA using $\alpha = 0.05$ is equivalent to a 39-bar simple moving average. A 39-day simple moving average has a 19-day lag, approximately half its length. Examination of Figure 2.3 shows that the very low-frequency lag of an EMA whose $\alpha = 0.05$ is indeed 19 days. Although the lag decreases as frequency is increased, it is of little consequence because

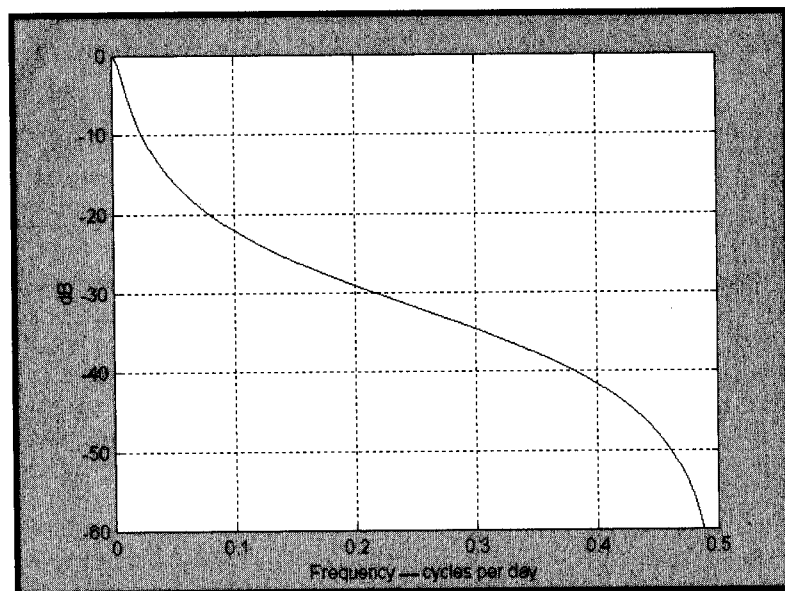


FIGURE 2.2 Smoothed-Input EMA Frequency Response ($\alpha = 0.05$)

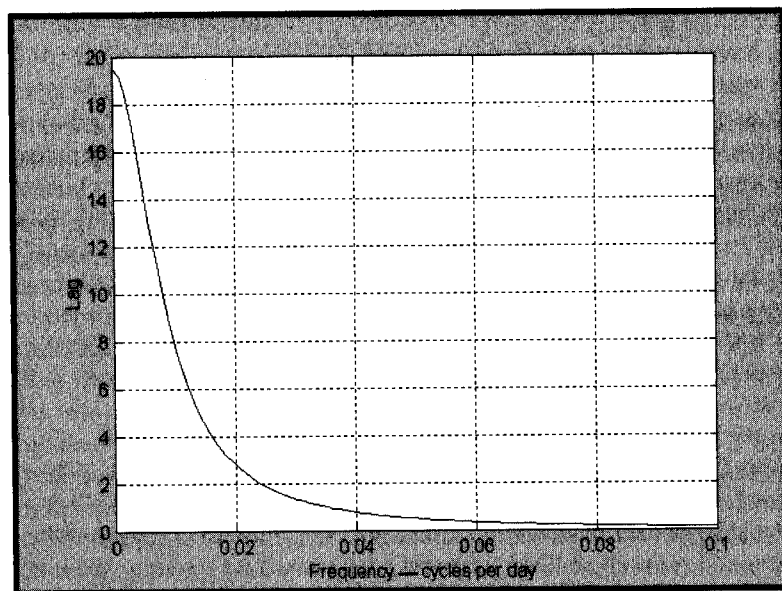


FIGURE 2.3 Smoothed-Input Lag Response ($\alpha = 0.05$)

the filtered amplitude is so small at these frequencies. The real impact of lag of all moving averages is the value of the lag at very low frequencies.

With Equation 2.3 we now have the capacity to construct a high-pass filter. We will subtract Equation 2.3 from unity as

$$\begin{aligned}
 HP(z) &= 1 - \frac{\frac{\alpha}{2} * (1 + Z^{-1})}{1 - (1 - \alpha) * Z^{-1}} \\
 &= \frac{1 - (1 - \alpha) * Z^{-1} - \frac{\alpha}{2} * (1 + Z^{-1})}{1 - (1 - \alpha) * Z^{-1}} \\
 &= \frac{\left(1 - \frac{\alpha}{2}\right) * (1 - Z^{-1})}{1 - (1 - \alpha) * Z^{-1}}
 \end{aligned} \tag{2.5}$$

Sharper attenuation can be obtained by using higher-order filters. However, I have learned that higher-order filters not only have greater lag, but they also have transient effects that impress false artifacts on their outputs. This is somewhat like ringing a bell: The ringing is more a function of the bell itself rather than a filtered response of a driving force. A reasonable compromise is the use of a second-order Gaussian filter. A second-order Gaussian low-pass filter can be generated by taking an EMA and immediately taking another identical EMA of the first EMA. This can be represented by squaring the transfer response. We can therefore obtain a second-order Gaussian high-pass filter response by squaring Equation 2.5 as

$$HP(z) = \frac{\left(1 - \frac{\alpha}{2}\right)^2 * (1 - 2 * Z^{-1} + Z^{-2})}{1 - 2 * (1 - \alpha) * Z^{-1} + (1 - \alpha)^2 * Z^{-2}} \tag{2.6}$$

Equation 2.6 is converted to an EasyLanguage statement as

$$\begin{aligned}
 HPF &= (1 - \alpha/2)^2 * (\text{Price} - 2 * \text{Price}[1] + \text{Price}[2]) \\
 &\quad + 2 * (1 - \alpha) * HPF[1] - (1 - \alpha)^2 * HPF[2];
 \end{aligned} \tag{2.7}$$

The transfer responses of Equations 2.6 and 2.7 (they are the same) are plotted in Figure 2.4.

Figure 2.4 shows that only frequency periods longer than 40 bars (frequency = 0.025 cycles per day) are significantly attenuated. Thus we have created a high-pass filter with a relatively sharp cutoff response. Since the output of this filter contains essentially no trending components, it must be the cycle component of price.

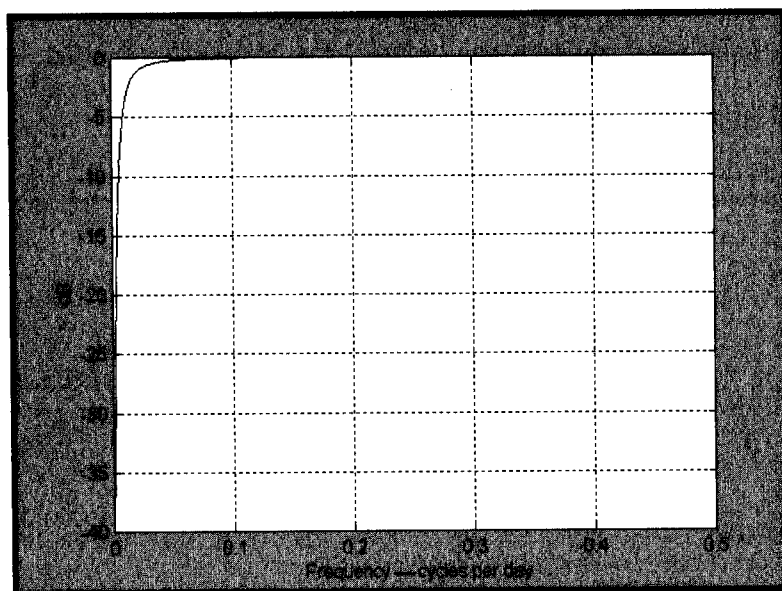


FIGURE 2.4 Transfer Response of a Second-Order High-Pass Gaussian Filter ($\alpha = 0.05$)

The complementary low-pass filter that produces the Instantaneous Trendline is found by subtracting the high-pass components of Equation 2.6 from unity. Skipping over the tedious algebra to put both elements of this subtraction over a common denominator, the equation for the low-pass Instantaneous Trendline is

$$IT(z) = \frac{\left(\alpha - \frac{\alpha^2}{4}\right) + \frac{\alpha^2}{2}Z^{-1} - \left(\alpha - \frac{3\alpha^2}{4}\right)Z^{-2}}{1 - 2 * (1 - \alpha) * Z^{-1} + (1 - \alpha)^2 Z^{-2}} \quad (2.8)$$

Equation 2.8 is converted to an EasyLanguage statement as

$$\begin{aligned} \text{InstTrend} = & (\alpha - (\alpha/2)^2) * \text{Price} + (\alpha^2/2) * \text{Price}[1] \\ & - (\alpha - 3\alpha^2/4) * \text{Price}[2] + 2 * (1 - \alpha) \\ & * \text{InstTrend}[1] - (1 - \alpha)^2 * \text{InstTrend}[2]; \end{aligned} \quad (2.9)$$

Figure 2.5 shows the attenuation of the Instantaneous Trendline filter and how only the low-frequency components are passed. The attenuation characteristic of the Instantaneous Trendline in Figure 2.5 is almost identical to that of the EMA shown in Figure 2.2.

The most important feature of the Instantaneous Trendline is that it

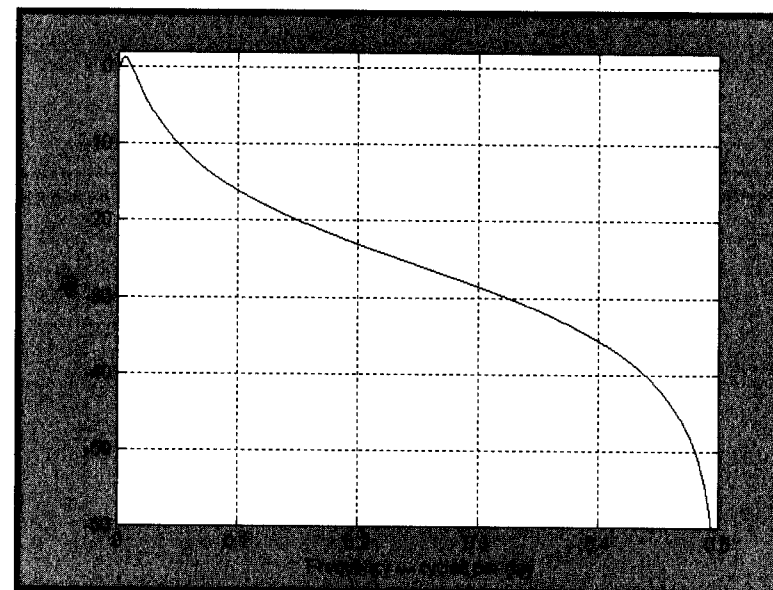


FIGURE 2.5 Frequency Response of the Instantaneous Trendline Filter ($\alpha = 0.05$)

has zero lag. That's right—*zero lag*! The lag is 0 because Instantaneous Trendline was created by subtracting the transfer response of a high-pass filter from unity. Since the high-pass filter has a very small amplitude at low frequencies, the resulting low-frequency lag of the difference is just the lag of unity, which is 0. Figure 2.6 shows the lag profile of the Instantaneous Trendline as a function of frequency. While the lag does increase to 13 bars at an approximate frequency of 0.005 cycles per day (200-day period), a frequency that low is more important to investors than to traders.

The importance of the zero lag feature of the Instantaneous Trendline is demonstrated by comparing its response to an EMA having an equivalent alpha. Figure 2.7 gives this comparison in response to real market data. It is clear that the two averages have about the same degree of smoothing, but that the Instantaneous Trendline has zero lag. If it is more convenient, you can think of the Instantaneous Trendline as a centered moving average. The major advantage of the Instantaneous Trendline compared to the centered moving average is that it can be used up to the right edge of the chart. That means that real indicators and trading systems can be built using it as a component. It is also clear that the lag of the Instantaneous Trendline is so small that a trader can begin to think about creating indicators and trading systems as a function of the price crisscrossing it. In later chapters we will develop such indicators and trading systems.

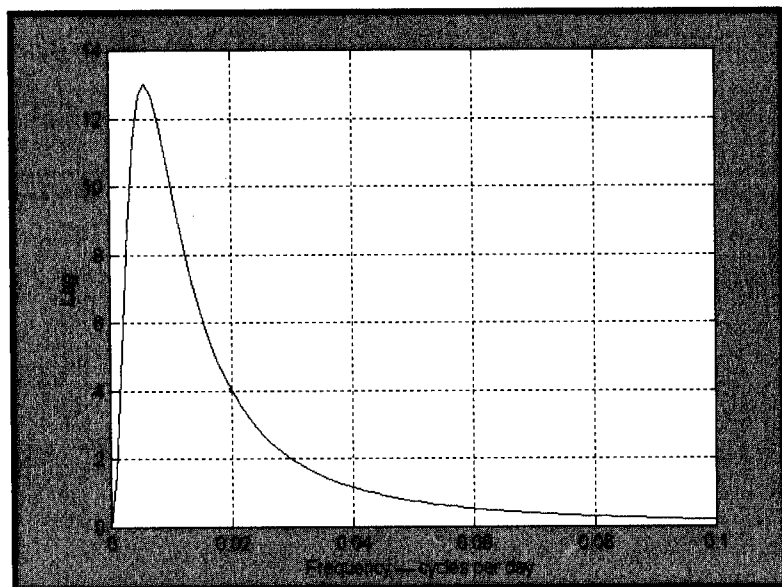


FIGURE 2.6 Lag of the Instantaneous Trendline Filter ($\alpha = 0.05$)

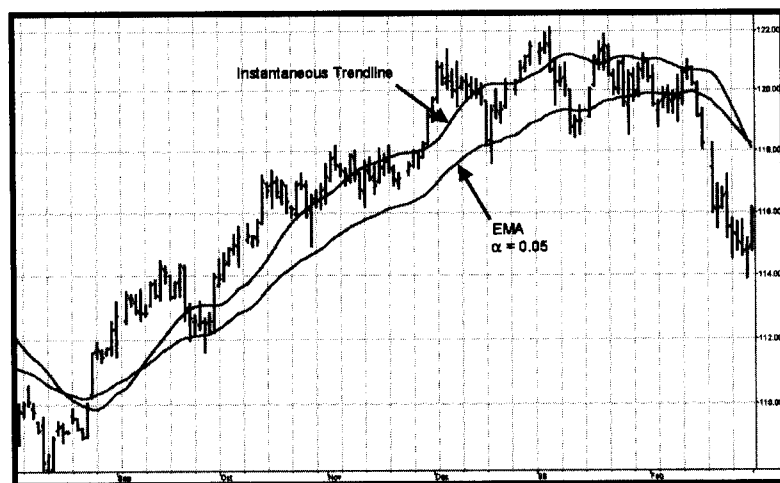


FIGURE 2.7 Instantaneous Trendline Has Much Less Lag than an EMA ($\alpha = 0.05$)

KEY POINTS TO REMEMBER

- The Instantaneous Trendline has zero lag.
- The Instantaneous Trendline has about the same smoothing as an EMA using the same alpha.
- An EMA is a low-pass filter.
- Higher-order Gaussian filters are the equivalent of applying the EMA multiple times.
- Using filters higher than second order is not advisable because of the ringing transient responses of the higher-order filters.
- A complementary cycle oscillator to the Instantaneous Trendline exists as a second-order high-pass filter.
- The lag of the complementary cycle oscillator is 0.